Somaiya Vidyavihar University
K J Somaiya School of Engineering

Batch: **Roll No.:**
Experiment / assignment / tutorial No. 02
Grade: AA / AB / BB / BC / CC / CD /DD
Signature of the Staff In-charge with date

Title: Linear Block Code
Objectives:
1. **To design encoder and decoder using Linear Block Code**

OUTCOME: apply different methods of error control coding and schemes required for reliable transmission of digital communication

Problem statement:

- **Design theoretically a (6, 3) linear block encoder for given the generator matrix and implement the circuit of the encoder and decoder**

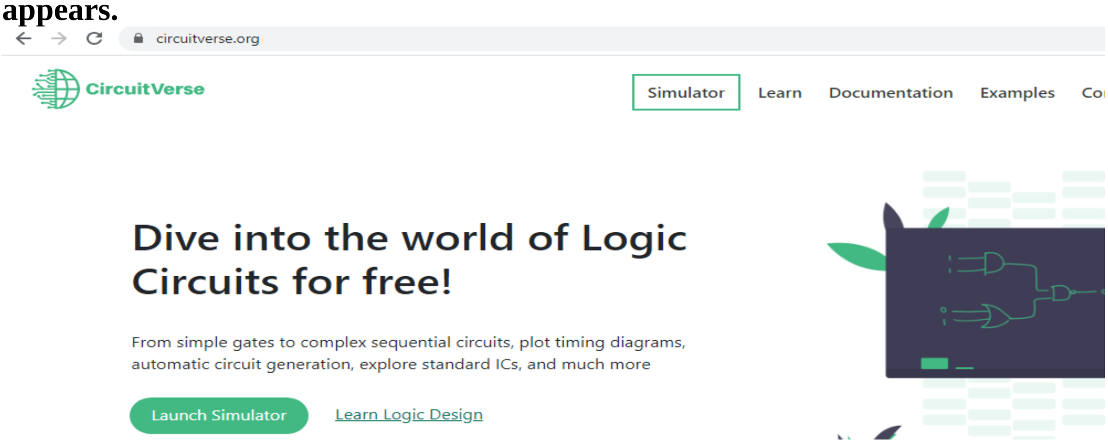
$$G = \begin{matrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{matrix}$$

Procedure:

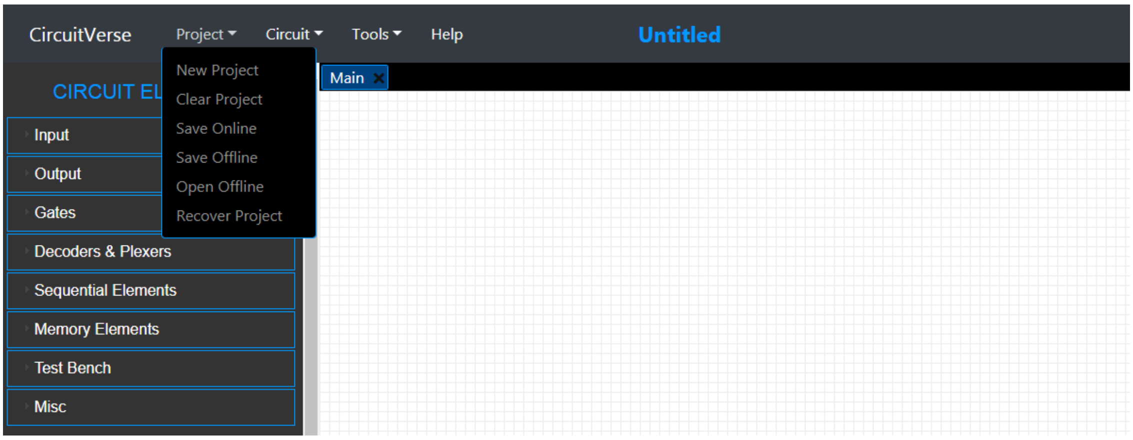
- **Design a digital circuit to implement the above encoder.**

Use open source simulator software: Circuit Verse (Just signup once or login with Google). (You may use LTpsice)

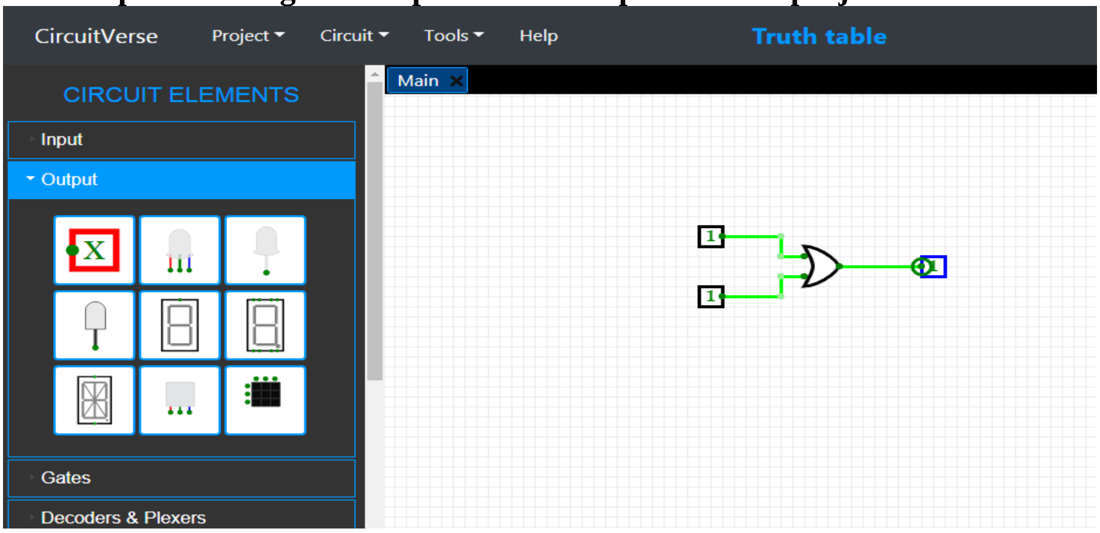
Step 1: Click on the option of “Launch Simulator”, the untitled project window appears.



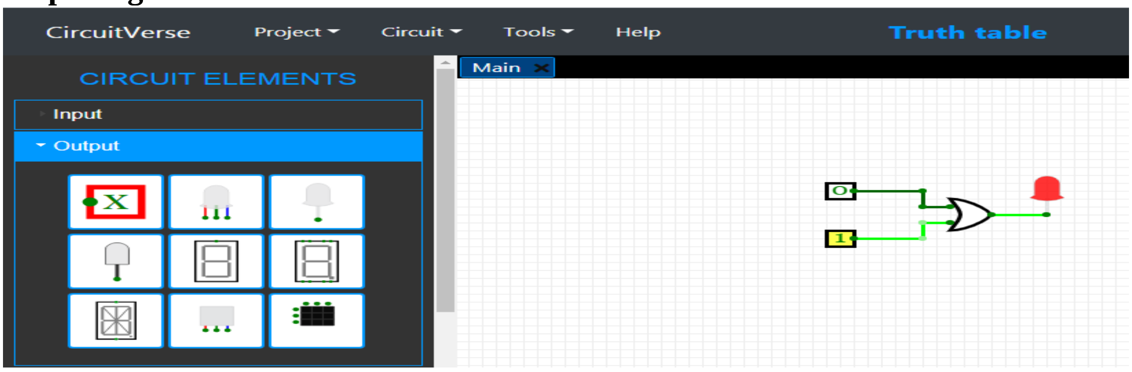
Step 2: Select the project tab of the toolbar and create a new project. Save the project in online mode.



Step 3: Select the appropriate input, logic gate and output option from the circuit element palette. Drag and drop the selected option on the project screen.



Step 4: Interconnect the placed elements by clicking on the two end-points. The output changes can be visualised in real-time. Note down the corresponding input-output logic



Step 5: Observe the code words for all the inputs. Verify using 'encode' function of MATLAB.

Step 6: Design a circuit (decoder) to evaluate the syndrome from the received code word

Step 7: Implement circuit, using software. Generate syndrome table using 'syndtable' function of MATLAB.

Step 8: Introduce single bit error in one code word. Evaluate the syndrome for it. Identify the error pattern. Verify results using hardware and software.

Expected Output:

- **Corresponding code vectors**
- **Minimum hamming distance d_{min}**
- **Error detection and error correction capability**
- **Parity check matrix**
- **Find error if received code is 1 0 0 0 1 1**

Discussion/Conclusion:

Signature of faculty in-charge